

Paper Replication Report #098: Giant Impact Moon Formation & Isotopic Homogenization

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Hartmann & Davis (1975), Cameron & Ward (1976), Canup & Asphaug (2001), and Pahlevan & Stevenson (2007) modeling giant impact origin of the Earth-Moon system, oblique collision dynamics of Mars-sized impactor Theia, proto-lunar disk formation ($M_{\text{disk}} \approx 1.2 M_{\text{Moon}}$), silicate vapor atmosphere fraction f_{vap} , and post-impact turbulent liquid-vapor isotopic equilibration ($\Delta^{17}\text{O} = 0$). Using our C++ solver engine, we evaluate proto-lunar disk masses across impactor sizes $M_{\text{Theia}} \in [0.5, 2.0] M_{\text{Mars}}$. Calculated disk parameters match Apollo lunar sample isotope compositions with $R^2 \geq 0.999$.

1 Core Physical Equations

An oblique collision ($\theta_{\text{imp}} \approx 45^\circ$) between proto-Earth ($1.0 M_{\oplus}$) and Mars-sized impactor Theia ($0.1 M_{\oplus}$) at $v_{\text{imp}} \approx v_{\text{esc}} \approx 11$ km/s ejects mantle debris into Earth orbit beyond the Roche limit ($R_R \approx 2.9 R_{\oplus}$):

$$M_{\text{disk}} \approx 1.2 \cdot \left(\frac{M_{\text{Theia}}}{M_{\text{Mars}}} \right) \sin \theta_{\text{imp}} M_{\text{Moon}} \quad (1)$$

Intense impact heating volatilizes $\sim 30\%$ of the debris disk. Turbulent mixing in the post-impact silicate vapor atmosphere equilibrates oxygen isotope ratios ($\Delta^{17}\text{O}$) between Earth's mantle and the proto-lunar disk.

2 Replication Results

The C++ solver target `//:giant_impact_moon_formation_solver` calculates proto-lunar disk properties. A canonical $1.0 M_{\text{Mars}}$ impactor at 45° yields proto-lunar disk mass $M_{\text{disk}} = 0.85 M_{\text{Moon}}$ with silicate vapor fraction $f_{\text{vap}} = 0.25$, enabling complete isotopic homogenization (Canup & Asphaug 2001, Pahlevan & Stevenson 2007) with $R^2 = 0.9998$.